

- 2 Nagarahole National Park in southern India is a forest ecosystem that is important for the conservation of large mammals such as the Asian elephant, *Elephas maximus*. Conflicts between people and elephants in the area surrounding the national park can threaten conservation work.

Fig. 2.1 shows a group of Asian elephants.



Fig. 2.1

Several coffee farms (farms where coffee is the crop) are found close to Nagarahole National Park. Asian elephants sometimes leave the national park and enter the coffee farms. Elephants damage crops and trees, and sometimes injure people who work on the farms.

Decender trees, *Erythrina subumbrans*, are planted on coffee farms to provide shade and are often damaged when elephants enter the farms.

Some scientists wanted to investigate the reasons why elephants enter coffee farms close to Nagarahole National Park.

20 coffee farms in an area to the south of Nagarahole National Park were chosen at random.

In March 2008, the scientists measured four variables at each coffee farm, as shown in Table 2.1.

Table 2.1

variable	how the variable was determined for each coffee farm
water availability	counting the number of ponds, lakes and water holes visible in satellite images
density of December trees	counting the number of December trees in ten circular areas, each with a radius of 10 m
grass biomass	drying and measuring the dry mass of all the grass leaves collected in 50 randomly selected areas of 0.25 m ²
distance from Nagarhole National Park	using GPS to measure the distance between the boundary of Nagarhole National Park and the boundary of the farm

Farmers from each coffee farm completed a questionnaire to record when elephants entered the farm.

The scientists decided to use Spearman's rank correlation to investigate the relationship between each of the four variables in Table 2.1 and the number of months that elephants entered each coffee farm per year.

- (a) State **two** conditions that allow Spearman's rank correlation to be used in this investigation.

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- (b) State a null hypothesis for testing whether there is a correlation between water availability and the number of months that elephants entered each coffee farm per year.

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- (c) Table 2.2 shows the results of using Spearman's rank correlation (r_s) to investigate the relationship between each of the four variables in Table 2.1 **and** the number of months that elephants entered each coffee farm per year. The table is **not** complete.

Table 2.2

variable	r_s value	significance
water availability	0.63	
density of December trees	– 0.30	
grass biomass	0.02	
distance to Nagarahole National Park	0.38	

Table 2.3 shows some critical values for the Spearman's rank correlation at different probabilities. When comparing critical values in Table 2.3 with values of Spearman's rank correlation that are negative, the minus sign should be ignored.

Table 2.3

number of paired observations	probability			
	0.50	0.10	0.05	0.01
18	0.170	0.401	0.472	0.600
19	0.165	0.391	0.460	0.584
20	0.161	0.380	0.447	0.570
21	0.156	0.370	0.435	0.556

- (i) For each variable in Table 2.2, decide whether the r_s value is **significant** or **not significant**. Record your decision in the third column of Table 2.2, headed **significance**. Write **yes** if the value is significant or **no** if the value is **not** significant. [1]
- (ii) Describe what can be concluded from Table 2.2 about the relationship between the density of December trees and the number of months that elephants entered each coffee farm per year.

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (d) A coffee farmer living close to a different national park in India wanted to reduce the number of times that elephants entered the farm.

The farmer analysed data from the investigation and concluded that reducing the number of ponds, lakes and water holes on the farm would reduce the number of times that elephants entered the farm.

Explain how the information provided **and** the data in Table 2.2 support and do **not** support this conclusion.

support

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do **not** support

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[3]

[Total: 8]